

Learning, Scarcity, and Fluid Benchmarks

Show your reasoning, not only the final answer. Unless a problem says otherwise, all random variables introduced in different periods or experiments are independent. You may use Hoeffding's inequality in the following form: if $X_1, \dots, X_n$ are iid random variables in $[0, 1]$ with mean μ and $\widehat{M}_n = n^{-1} \sum_{s=1}^n X_s$, then for every $a > 0$,

$$\mathbb{P}(\widehat{M}_n - \mu \geq a) \leq e^{-2na^2}, \quad \mathbb{P}(\mu - \widehat{M}_n \geq a) \leq e^{-2na^2}.$$

Problem 1: A/B testing and confidence bounds (30 points)

Rewards from every policy lie in $[0, 1]$. For policy i , write $X_{i,s}$ for the random reward on its s th test, $\mu_i = \mathbb{E}[X_{i,s}]$ for its fixed but possibly unknown mean, and

$$\widehat{M}_{i,n} = \frac{1}{n} \sum_{s=1}^n X_{i,s}$$

for its random empirical mean. Fix a practically meaningful improvement $\Delta > 0$ and an error tolerance $\delta \in (0, 1)$.

- (a) **One challenger; known incumbent mean.** (10 points) An incumbent has known mean μ_0 . A challenger with unknown mean μ_1 is tested independently n times. You are promised that one of the separated hypotheses holds:

$$H_0 : \mu_1 \leq \mu_0, \quad \text{or} \quad H_1 : \mu_1 \geq \mu_0 + \Delta.$$

Propose a decision rule based on $\widehat{M}_{1,n}$. Use Hoeffding's inequality to bound the probability of an incorrect decision under each hypothesis, and give a sufficient sample size that makes each error probability at most δ . Interpret how the required sample size depends on Δ and δ .

- (b) **Many challengers.** (12 points) There are K challengers, each tested n times, and the incumbent mean μ_0 is known. You are promised that either

$$H_0 : \mu_i \leq \mu_0 \quad \text{for every } i = 1, \dots, K,$$

or

$$H_1 : \mu_i \geq \mu_0 + \Delta \quad \text{for at least one } i.$$

Under H_1 , the remaining challengers may have arbitrary means in $[0, 1]$. Design a fixed-sample procedure and bound both the probability of a false positive and the probability of missing every genuinely better challenger. Find a sufficient number of tests per challenger and the resulting total challenger-sample budget. Identify exactly where the dependence on K enters. Why might allocating the same number of tests to every challenger be wasteful? Briefly relate your answer to UCB or successive elimination; no regret proof is required.

- (c) **The incumbent is estimated too.** (8 points) Now μ_0 is unknown. The incumbent has already been tested N_0 times, and one challenger is tested n times. The same two separated hypotheses from part (a) apply. Construct simultaneous confidence intervals for μ_0 and μ_1 , and use them to give a reliable midpoint-threshold decision rule. Derive a sufficient condition involving N_0, n, Δ , and δ that makes the classification error at most δ under either hypothesis.

Explain why arbitrarily many challenger tests cannot compensate for an extremely imprecise incumbent estimate.

You may use the confidence radius

$$r(m, \varepsilon) = \sqrt{\frac{\log(2/\varepsilon)}{2m}}.$$

Be careful about strict inequalities at the decision threshold.

Problem 2: Littlewood's rule (20 points)

An airline has capacity $C = 10$. Low-fare customers arrive first and pay $r_L = 100$; high-fare customers arrive later and pay $r_H = 250$. High-fare demand is the integer-valued random variable

$$D_H \sim \text{Binomial}(8, 1/2).$$

Low-fare demand is sufficiently abundant to fill every seat not protected for high-fare demand. The airline chooses an integer protection level $y \in \{0, 1, \dots, C\}$: it sells $C - y$ seats to low-fare customers and reserves y seats for the later high-fare demand.

- (a) (4 points) Write expected revenue $R(y)$ as a function of y .
- (b) (4 points) Compute $R(y) - R(y - 1)$ for $y \geq 1$ and explain the event on which protecting the y th seat creates high-fare revenue.
- (c) (6 points) Derive Littlewood's marginal comparison and compute an optimal protection level for this instance. State your tie-breaking convention and use the correct discrete tail probability.
- (d) (2 points) Convert the protection level into a low-fare booking limit.
- (e) (4 points) Without recomputing the binomial table, explain what happens to the optimal protection level if (i) r_H increases, or (ii) high-fare demand becomes stochastically larger, meaning that all its tail probabilities weakly increase. Your answers may be weak comparative statics because ties can occur.

Problem 3: Dynamic capacity allocation (25 points)

There are T periods and initial capacity C . In each period, independently, a high-fare request arrives with probability q_H , a low-fare request arrives with probability q_L , and no request arrives with probability $q_0 = 1 - q_H - q_L$. Assume $q_H, q_L \geq 0$ and $q_H + q_L \leq 1$. The seller observes the request class before deciding whether to accept it. An accepted request uses one unit of capacity and earns r_H or r_L , where $0 < r_L < r_H$; rejected demand is lost.

Let $v_t(b)$ be the optimal expected revenue with t periods and b units of capacity remaining.

- (a) (3 points) State the boundary conditions.
- (b) (6 points) Derive the Bellman recursion. You may take as given that a high-fare request is accepted whenever capacity is available.
- (c) (4 points) Express the decision to accept a low-fare request using the marginal continuation value

$$v_{t-1}(b) - v_{t-1}(b - 1).$$

Specify what your policy does at equality.

- (d) (6 points) Solve the instance

$$T = 3, \quad C = 2, \quad r_H = 10, \quad r_L = 4, \quad q_H = \frac{1}{2}, \quad q_L = \frac{1}{4}$$

by tabulating $v_t(b)$ for $t = 0, 1, 2, 3$ and $b = 0, 1, 2$. For every state with a low-fare arrival, indicate whether the optimal action is accept or reject under your tie-breaking convention.

- (e) (3 points) Use your table to explain why a fixed protection decision need not be optimal when the two fare classes are interleaved.
- (f) (3 points) Define an offline clairvoyant who sees the full arrival sequence before making any decisions. Explain why its expected revenue upper-bounds $v_T(C)$.

Problem 4: Jensen's inequality and fluid bounds (25 points)

Throughout this problem, the demand random variables have finite support.

- (a) **Finite Jensen.** (5 points) Let g be concave and let X take values $x_1, \dots, x_n$ with probabilities $\theta_1, \dots, \theta_n$. Starting from the two-point definition of concavity, prove by induction that

$$\mathbb{E}[g(X)] \leq g(\mathbb{E}[X]).$$

- (b) **One random demand.** (6 points) Capacity is C , high-fare demand is D_H , and low-fare demand is abundant enough to fill all capacity not used by high-fare demand. Define the offline revenue from realized high-fare demand $d \geq 0$ by

$$g(d) = r_H \min\{d, C\} + r_L(C - d)^+, \quad r_H > r_L > 0.$$

Simplify $g(d)$ separately for $d \leq C$ and $d \geq C$, prove that g is concave, and use part (a) to show

$$\mathbb{E}[g(D_H)] \leq g(\mathbb{E}[D_H]).$$

Interpret this inequality economically.

- (c) **Two-class fluid LP.** (9 points) Now both class demands can be random. For deterministic demand bounds $d = (d_H, d_L) \in \mathbb{R}_+^2$, define

$$\begin{aligned} \text{FLUID}(d_H, d_L) = \max_{x_H, x_L \geq 0} \quad & r_H x_H + r_L x_L \\ \text{s.t.} \quad & x_H + x_L \leq C, \\ & x_H \leq d_H, \quad x_L \leq d_L. \end{aligned}$$

Show that $\text{FLUID}(d)$ is concave in the demand-bound vector d . Then use Jensen's inequality to prove

$$\mathbb{E}[\text{OPT}(D_H, D_L)] \leq \text{FLUID}(\mathbb{E}[D_H], \mathbb{E}[D_L]),$$

where $\text{OPT}(D_H, D_L)$ is the realized revenue of an offline clairvoyant.

Hint for concavity: take optimal feasible allocations for two demand vectors. What can you say about a convex combination of those allocations?

- (d) **Three benchmarks.** (5 points) Explain the two inequalities in

$$\text{FLUID}(\mathbb{E}[D_H], \mathbb{E}[D_L]) \geq \mathbb{E}[\text{OPT}(D_H, D_L)] \geq v_T(C).$$

Identify which inequality comes from Jensen's inequality and which comes from the extra information available to the offline clairvoyant.

Optional extension (ungraded): computation at scale

Implement the two-fare dynamic program from Problem 3 for increasing horizons T , with $C = [0.55T]$, $q_H = 0.35$, $q_L = 0.40$, $r_H = 10$, and $r_L = 4$. Compare $v_T(C)$ with the fluid benchmark $\text{FLUID}(Tq_H, Tq_L)$. Plot both the absolute gap and the gap divided by $\sqrt{T}$ for a sensible range of T . Describe what you observe, but do not infer a universal asymptotic theorem from the numerical plot.